



Figure 1: Manual counter designer screen

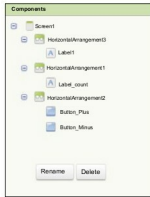


Figure 2: Manual counter components

- Both buttons have been put within the 'Horizontal arrangement', so they are aligned horizontally.
- We have also changed the background colour property of both buttons to identify them more clearly.
- If you have dragged and placed everything, the layout will look somewhat as in Figure 1.
- The components viewer will look like what's shown in Figure 2.
- Make the necessary property changes like we did earlier when changing the text property, for the label and button components.
- So this way, your graphical user interface is ready. Figure 1 shows exactly what the application will look like after installation.

Before heading towards the blocks editor to define the behaviour, let's discuss what actual functionality we expect from our application.

- When we press the 'Plus' button, the counter value should increase.
- When we press the 'Minus' button, the counter value should decrease.

So let's move on and add these two features using the block editor. I hope you remember how to switch from the designer to the block editor. There is a button available right above the 'Properties' pane to switch between the two.

Block editor blocks

- The blocks shown in Figure 3 are easy to understand. These indicate that when the button 'Plus' is pressed, it should increase whatever figure is shown as the count label.
- When the button 'Minus' is pressed, it should decrease whatever figure is shown as the count label.
- The additional 'Plus' and 'Minus' blocks you see fall under the mathematics category.

The automatic counter

In the last section, you counted things manually using buttons. You pressed the 'Plus' or 'Minus' buttons to get the desired result. You will now be able to build another useful app,

which is a simple countdown timer that will be triggered on pressing a button and will show you the time elapsed in seconds, minutes or hours.

GUI requirements

Not much is needed by way of GUI requirements as the whole thing is very basic. We will require two button components and a label component, along with some horizontal arrangements to keep things aligned.

We will require the following components for this project:

	Component name	Purpose	Location
1	Horizontal arrangement	To arrange the child components horizontally	Palette--> Layout--> Horizontal Arrangement
2	Label	To display the count value	Palette-->User Interface-->Label
3	Button	One button to increase the count	Palette-->User Interface-->Button
4	Button	One button to decrease the count	Palette-->User Interface-->Button
5	Clock	To count time	Palette-->Sensors-->Clock

- Drag and drop the components mentioned above to the viewer.
- Visible components can be seen, while the non-visible components will be located beneath the viewer under the tag 'Non-visible'.
- We have placed a label to put the name of the application.
- All buttons have been put within the 'Horizontal arrangement' so as to keep them aligned horizontally.
- We have also changed the background colour property of both buttons to identify them more easily.
- You will see the clock component for the very first time. It has access to the inbuilt clock of the phone and performs actions accordingly.
- If you have dragged and placed everything, the layout will look something like what's shown in Figure 4. Figure 5 shows what the components viewer will look like. Since you are not very familiar with the clock component and its properties, set the properties as depicted in the picture shown in Figure 6.

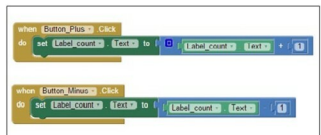


Figure 3: Manual counter block editor blocks